CHAPTER 2 · CONCEPT 1 · WORKSHEET

Exponential Notation — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 · MCQ **[1]**

In 5^4, the number 4 is called the:

(a) base (b) exponent (c) factor (d) product

Q2 · MCQ **[1]**

648 expressed as prime powers is:

(a) $2^3 \times 3^4$ (b) $2^4 \times 3^3$ (c) $2^3 \times 3^3$ (d) $2^2 \times 3^4$

Q3 · FILL IN THE BLANK **[1]**

The value of $(-2)^4$ is:

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Q4 · MCQ **[1]**

The value of -2^4 (no brackets) is:

(a) 16 (b) −16 (c) 8 (d) −8

Q5 · TRUE / FALSE **[1]**

$(-5)^3$ is a negative number.

True False

Q6 · ASSERTION–REASON [1]

Assertion (A): $(-3)^2 \times (-5)^2 = 225$.
Reason (R): A negative number raised to an even power is positive.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7 [2]

Express in exponential form: (i) $2 \times 2 \times a \times a$ (ii) $5 \times 5 \times 7 \times 7 \times 7$

Q8 [2]

Express 405 and 540 as products of their prime factors in exponential form.

Q9 [2]

Find the numerical value of $3^2 \times 10^4$.

Long answer (3 marks each)

Q10 [3]

Express each as a product of powers of prime factors in exponential form: (i) 648 (ii) 405 (iii) 540

Q11 [3]

Write the numerical value of each: (i) $(-3)^2 \times (-5)^2$ (ii) $3^2 \times 10^4$ (iii) $(-2)^5 \times (-10)^6$

Q12 [3]

A bacterium doubles every hour. Starting with 1 cell, how many cells are there after 8 hours? Express the answer as a power of 2 and as an ordinary number.

Total: 21 marks · Check your work against the answer key.